

Topological Memory - Dependence

David Ponarovsky

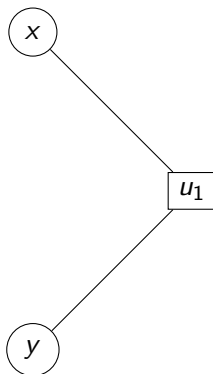
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Introduction

Today:

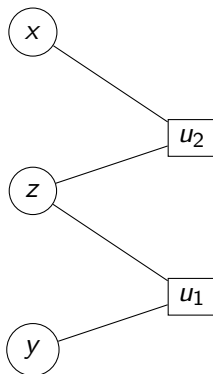
- ▶ Dependence.
- ▶ Local Stochastic Noise.

Dependence Type I



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\ &= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Dependence Type II



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\ &= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Definition

Consider two (qu)bits x, y . We say that z is a **linked node** between x and y if z is either a bit or a check that is shared by both of their D light cones.

Remark

Notice that y might be a **linked node** between x and y .

Definition

Consider an application of a noise channel. We say that z is a **clean linked node** if:

1. z is a bit and $z \notin f$ (not flipped).
2. z is a check and z is satisfied.

$$\begin{aligned}
& \Pr[x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_l \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z_i \in \Gamma(x_i)] \\
&\leq \prod_{i=1}^k \Pr[x_i \in e] \prod_{i=k+1}^l \Pr[x_i \in e \mid z_i \in \Gamma(x_i)] \\
&\leq M(p)^k
\end{aligned}$$

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.

Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits S , we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)